

Asma

Department of Linguistics, University of Arizona
asma1@gmail.com | Tucson, AZ

Education

Ph.D. Linguistics, University of Arizona, Tucson, AZ (Expected May 2027)
Major: Syntax | Minor: Computational Linguistics

M.S. Human Language Technology, University of Arizona (Expected Fall 2026)

M.Phil. Applied Linguistics, Riphah International University, Pakistan (2021)
Focus: SLA, Language Pedagogy, Stylistics and Discourse.

M.A. English Language Teaching and Linguistics, University of the Punjab, Pakistan (2019)

Research Interest

Syntax, Neurolinguistics, Natural Language Processing, South Asian Languages (Urdu, Punjabi, etc.)

Experience

Research Internship | University of Arizona | Tucson Jan 2026 – May 2026

- **Supervisor:** Eric Jackson
- Designed and carried out an applied NLP research project focused on a data-scarce languages of Pakistan
- Improved neural machine translation models by using CFG generated artificial data.

Instructor of Record | University of Arizona | Tucson Aug 2022 – Continue

- Independently designed and taught undergraduate courses, including syllabus creation, lectures, assignments, and assessments.
- **Courses Taught:**
 - Ling 432 – Psychology of Language (Summer 2026) - [syllabus](#)
 - Ling 341 – Language Development (Spring 2026) - [syllabus](#)
 - Ling 341 – Language Development (Spring 2025) - [syllabus](#)
 - Ling 432 – Psychology of Language (Fall 2024) - [syllabus](#)
 - Ling 150A1 - Introduction to Linguistics (Summer 2024) - [syllabus](#)
 - Ling 150A1 - Introduction to Linguistics (Summer 2023) - [syllabus](#)

Organizer, SynSalon | University of Arizona | Tucson

Spring 2025

- Host and coordinate SynSalon, a discussion & presentation series focused on syntax and related topics in linguistics.

Teacher | Azam Garrison | Lahore

Feb 2019 – Mar 2022

- Class teacher of Grade 5.
- Teaching English, Urdu and Math based on the pre-designed curriculum.
- Helping the class with annual projects and competitions.

Publications and Presentations:

Publication

- Asma. (Under review). *Volitionality Markers in Urdu*. Linguistic Inquiry.

Book Chapters

- Asma. (in press). *Key differences between neurolinguistics and psycholinguistics*. In A. Carnie (Ed.), *Linguistics: A modular introduction*. University of Arizona Press.
- Asma. (in press). *Language regions of the brain*. In A. Carnie (Ed.), *Linguistics: A modular introduction*. University of Arizona Press.
- Asma. (in press). *Aphasia*. In A. Carnie (Ed.), *Linguistics: A modular introduction*. University of Arizona Press.

Keynote Speaker:

- Asma and Carnie, Andrew. (2026). *Reconciling the differences between AI and Generative Grammar*. Riphah International University Conference on Language, Literature, and Culture (RICOLLAC). April 3-5.

Conference Presentation

- Asma. (2026). *Volitionality Markers in Punjabi*. Linguistic Society of America Annual Meeting, New Orleans, Louisiana. Jan 8-11.
- Asma. (2025). “*A Corpus-Based Description of Dawoodi Syntax*”. Arizona Linguistic Circle 18. University of Arizona.
- Asma. (2024). “*The Effect of Word Order and Semantic Distributivity on Agreement Attraction Errors in Urdu*”. The 15th Newcastle and Northumbria Postgraduate Conference in Language and Linguistics (NNPCiLL), Newcastle, UK.
- Asma. (2019) “*A Comprehensive Study of /h/-dropping in Urdu*”. The International Conference of Department English Language and Linguistics, University of Management and Technology, Lahore. (April 6, 2019)

Recent Projects

Volitionality & NLLB: A LKMT Approach for Urdu and Punjabi Translation

2026

University of Arizona – Research Internship, Fall 2026 | Machine Translation, Neural Networks, NLLB

- Fine-Tuned NLLB for specific sentences in Urdu and Punjabi that encode the presence and absence of volitionality through case markers by using a complex CFG generated artificial corpus.
- Improved accuracy from 20-30% to 70-80%.

Urdu Case Classifier

2026

Joint Project with Matthew Harnandez | Machine Learning, Neural Networks, Classification

- Created a Case Classifier for Urdu by using a complex artificially generated Urdu dependency grammar.
- The Urdu dependency grammar was created using CFG and rule mapping. This strategy helped us in training the classification model even in the absence of good quality data for this low resource language.

Sentiment Classification Neural Network

2025

University of Arizona – Neural Networks, Spring 2025 | Machine learning, Neural Networks, AI

- Developed and tuned a neural network for text sentiment classification as part of a competitive class project using Tensorflow.
- Achieved leaderboard evaluation by submitting formatted predictions on dev/test sets with version-controlled code.

Interactive Syntax Tree Drawing Tool

2022

University of Arizona - Computational Linguistics Course Project – Fall 2022 | HTML, JavaScript

- Built a web-based tool for drawing syntax trees without bracket notation, featuring an intuitive user-friendly interface.
- Enabled real-time tree rendering using JavaScript DOM manipulation to support linguistics education and research.

Awards & Acknowledgements**Peter W. Likins Graduate Fellowship** – University of Arizona, USA.

2023

- *Awarded for exceptional resilience and academic strength.*

Riphah Tuition Scholarship – Riphah International University, Pakistan.

2020

- *Granted for outstanding academic performance based on CGPA.*

Gold Medal – University of the Punjab, Pakistan.

2018

- *Awarded to the top-ranking student in the program for highest CGPA and academic excellence.*

High Achievers Scholarship – University of the Punjab, Pakistan.

2017

- *Granted for maintaining a consistently high academic performance.*

Grants**SBS Dissertation Grant** (\$3600) – University of Arizona, USA.

2026

- *Awarded for field-work and participant payment.*

GPSG Travel Grant (\$1000)– Department of Linguistics, University of Arizona, USA. 2025

- *Awarded for conference travel.*

Travel & Research Grant. (\$600) – Department of Linguistics, University of Arizona, USA. 2025

- *Awarded for conference travel.*

Travel & Research Grant (\$250) – Department of Linguistics, University of Arizona, USA. 2023

- *Awarded for subject compensation during research.*

Skills

- **Programming:** Python (TensorFlow, PyTorch), MATLAB, JavaScript, Git, HTML, Electron, R.
- **Brain Imaging & Psycholinguistics:** fMRI preprocessing, signal analysis, fMRI data collection, PsychoPy, Audacity.
- **Research:** Experimental design, data visualization, academic writing

Languages

- **Urdu** (First language (acquired); Native Proficiency)
- **Punjabi** (Second language (acquired); Native-like proficiency but slightly less fluent)
- **English** (Third language (learned); Good Proficiency. IELTS score of 8 (Speaking))